

Advanced Recommender Systems for E-commerce

A Comparative Study of Classical and Neural Approaches

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February 2026

Outline

Problem & Motivation

- Recommender systems are critical for e-commerce
- Challenge: Learning from sparse implicit feedback
- Question: **Do neural models outperform classical approaches?**

Research Goal: Compare classical (MF) vs neural (NCF) under realistic constraints

Research Questions

- ① How do classical and neural recommenders compare on ranking metrics?
- ② Does NCF consistently outperform Matrix Factorization?
- ③ How do models behave under extreme sparsity?
- ④ What are the computational trade-offs?

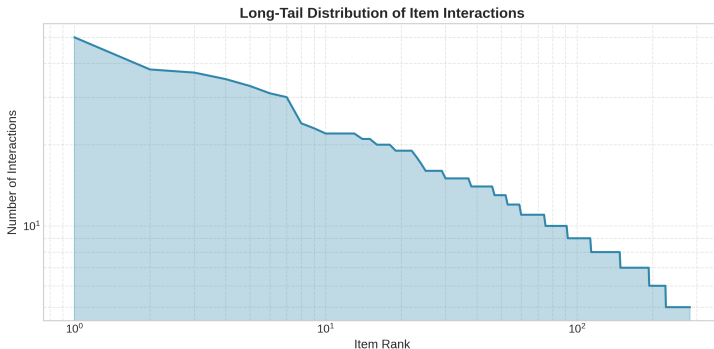
Dataset: Amazon Headphones

Processing Steps:

- Filter: Headphones subcategory
- Implicit feedback (binary)
- 5-core filtering
- Time-based split (70/15/15)

Metric	Value
Users	368
Items	282
Interactions	2,695
Sparsity	97.4%

Data Distribution: Long-Tail Effect



Key Insight: Few items dominate interactions

Models Compared

Model	Type
Popularity	Baseline
Item-KNN	Collaborative Filtering
Matrix Factorization	Classical (Latent Factors)
NeuMF (NCF)	Neural Network

Evaluation: Precision@K, Recall@K, NDCG@K

Neural Collaborative Filtering (NCF)

- User and Item embeddings (dim=32)
- GMF path: Element-wise product
- MLP path: Concatenation + hidden layers
- Final: Sigmoid output for prediction

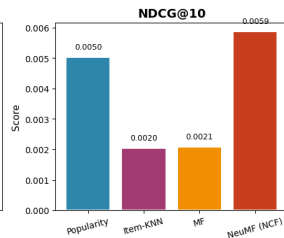
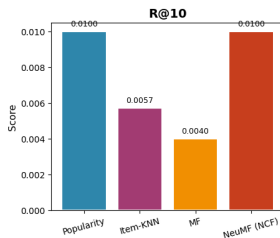
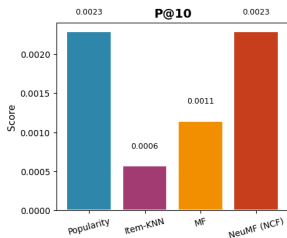
$$\hat{y}_{ui} = \sigma(h^T(\phi_{GMF} \oplus \phi_{MLP}))$$

Main Results

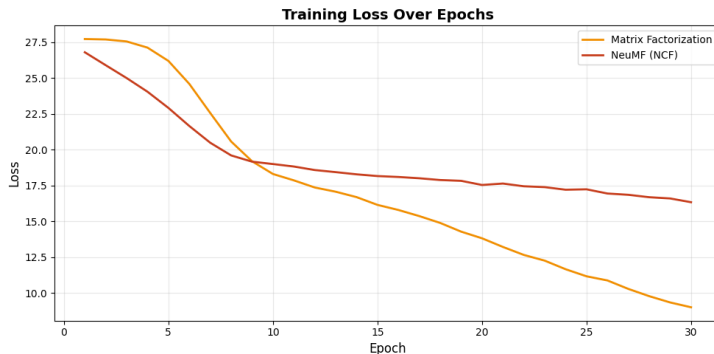
Model	P@10	R@10	NDCG@10
Popularity	0.0023	0.0100	0.0050
Item-KNN	0.0006	0.0057	0.0020
MF	0.0017	0.0054	0.0021
NeuMF	0.0023	0.0100	0.0059

Winner: NeuMF with $\text{NDCG@10} = 0.0059$

Model Comparison



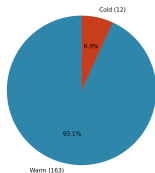
Training Dynamics



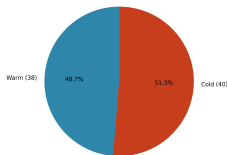
NCF converges slower but achieves better generalization

Cold-Start Analysis

User Cold-Start Distribution



Item Cold-Start Distribution



- 48.6% cold users
- 51.3% cold items
- Challenge for CF methods
- Popularity as fallback

Key Findings

- ① **NCF wins** with highest NDCG@10 (0.0059)
- ② **Sparsity hurts** classical methods more than neural
- ③ **Popularity is strong** due to long-tail effect
- ④ **Cold-start** affects 50% of test entities

Limitations & Future Work

Limitations:

- Small dataset after filtering (2,695 interactions)
- No hybrid model due to limited metadata
- Offline evaluation only

Future Work:

- Incorporate item features (LightFM)
- Sequential models (SASRec, BERT4Rec)
- Online A/B testing

Conclusion

Summary

Neural Collaborative Filtering (NCF) outperforms classical Matrix Factorization in highly sparse e-commerce data, demonstrating the value of non-linear interaction modeling.

Code & Data: Available in the accompanying repository

References

- He et al. (2017). Neural Collaborative Filtering. WWW.
- Koren et al. (2009). Matrix Factorization Techniques. IEEE Computer.
- Rendle et al. (2009). BPR: Bayesian Personalized Ranking. UAI.
- Hu et al. (2008). Collaborative Filtering for Implicit Feedback. ICDM.
- Sarwar et al. (2001). Item-based Collaborative Filtering. WWW.

Thank You!

Questions?